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The role of Doppler velocimetry in the management of high risk pregnancies

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ABSTRACT

Objective To determine whether knowledge of the result of Doppler velocimetry of the umbilical artery is beneficial to the management of a high risk pregnancy.

Design Randomised controlled trial. The trial was of the management type, designed to assess benefit accruing from additional information supplied by Doppler velocimetry.

Setting Tygerberg Hospital, Cape Town, South Africa. The hospital serves a population from the lower socio-economic groups.

Subjects Women with pregnancies 28 or more weeks gestation with hypertensive diseases and/or suspected small for gestational age fetuses were referred for Doppler velocimetry. From this population, three subsets were formed: 1. those with fetuses with absent end–diastolic velocities (20 fetuses); 2. those with hypertension but with fetuses with end–diastolic velocities (89 fetuses); and 3. those with fetuses suspected of being small for gestational age but with end–diastolic velocities (104 fetuses).

Interventions Doppler velocimetry on all subjects. The study group consisted of 10 cases with absent end–diastolic velocities, 47 cases with hypertensive diseases with end–diastolic velocities and 51 cases with suspected small for gestational age fetuses but with end–diastolic velocities in which the result was revealed to the clinician. The control group consisted of 10, 42 and 53 cases, respectively, in which the Doppler results were not revealed. All other routine investigations (sonar and antenatal fetal heart rate monitoring) were available to the clinicians. Standard management protocols were followed in all groups.

Main outcome measures Perinatal mortality and morbidity, antenatal hospitalisation, maternal intervention, admission to the neonatal intensive care unit and hospitalisation until discharge from the neonatal wards.

Results In the study and control groups the gestational age at entry to the study, maternal age, parity and various complications were not significantly different. In the subset with absent end–diastolic velocities, there was one neonatal death in the study group, but in the control group there were six deaths, five intrauterine and one perinatally related infant death ($P = 0{\cdot}029$). Because of this significant finding, the study was stopped. There were no differences in outcome in the subset where there was hypertensive disease with end–diastolic velocities between the study and control groups. In the subset in which small for gestational age fetuses were suspected, but in which end–diastolic velocities were present, the women in the study group had significantly fewer days in hospital before delivery ($P < 0{\cdot}001$) and tended to have fewer maternal interventions (study group = 27%, control group = 43%; $P = 0{\cdot}07$; odds ratio (OR) 0·49, 95% confidence limits (CL) 0·2 and 1·25) and caesarean sections (study group = 13%, control group = 27%; $P = 0{\cdot}08$; OR 0·43, 95% CL 0·14 and 1·32). The infants of the study group in this subset also spent significantly less time in the neonatal wards ($P = 0{\cdot}029$).

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Conclusions Within the confines of this study, knowledge of the Doppler velocimetry result was beneficial in the subsets with absent end–diastolic velocities and in which intrauterine growth retardation was suspected. In the subset in which women had hypertension but whose fetuses had end–diastolic velocities, there was no beneficial or adverse effect.

At a National Institute of Health consensus conference in 1986 on Doppler ultrasound (Low 1991), it was emphasised that even if altered velocity waveforms were confirmed as being able to detect fetal compromise with good sensitivity and specificity, additional clinical trials would be necessary to determine whether intervention based on Doppler ultrasound findings actually improved outcome. Neilson (1987) also urged caution on using Doppler ultrasound in clinical management until benefits for mother or fetus had been proven by the appropriate trials. In a review of published studies, Low (1991) concluded 'that no evidence had emerged during the last five years to support routine use of maternal or fetal blood flow velocity determinations as a diagnostic test in the general obstetric population'. Knowledge gained from the use of Doppler velocimetry of the umbilical artery (henceforth referred to as Doppler velocimetry) will only be of value if it alters the clinician's management so that the woman and/or her baby benefits.

The knowledge gained from Doppler velocimetry appeared to us to be of potential value mainly in two circumstances. Firstly, it is most likely to be of value in those fetuses where absent end–diastolic velocities (AEDV) of the umbilical artery are detected. This is not only because AEDV detected by Doppler velocimetry is associated with a very high perinatal mortality (McParland *et al.* 1990; Pattinson *et al.* 1993), but also and more importantly because changes in the Doppler indices usually precede abnormal fetal heart rate patterns (Pattinson *et al.* 1991). The awareness that fetuses demonstrating AEDV are at high risk of developing abnormal fetal heart rate patterns and dying may make the clinical management of the pregnancy more aggressive, thereby improving the fetal outcome provided the fetus is viable. In our population, AEDV occurs most frequently in cases of severe maternal hypertension or where severe small for gestational age fetuses are suspected.

Secondly, knowledge gained from Doppler velocimetry may be of use in the management of fetuses considered small for gestational age but where end–diastolic velocities (EDV) of the umbilical artery are present. This is because fetuses suspected of being small, but having normal Doppler indices, largely have been found to have a benign outcome (Rochelson *et al.* 1987; Haddad *et al.* 1988; Burke *et al.* 1990; Pattinson 1992); a more conservative management of this group may decrease the intervention rate in the women. It is possible that most are simply small, but normal, babies.

Therefore, Doppler velocimetry mainly may be of use in a selected group of women, namely those with hypertensive diseases and/or those where a small for gestational age baby is suspected. It was decided to perform a randomised controlled trial of management in this group of women to test whether the knowledge gained from Doppler velocimetry is beneficial to the management of these pregnancies.

Subjects and methods

A randomised controlled trial of the management type (which tests the benefit accruing from the additional information supplied by the Doppler velocimetry) was designed. Pregnant women with hypertensive disorders, or who were suspected of having small for gestational age fetuses, and who were referred for Doppler examinations were invited to take part in the trial. All obstetricians at our institution agreed to the trial provided their patients consented.

Three randomisation groups were created based on the clinical picture and Doppler velocimetry result, namely, an AEDV group comprising all fetuses with AEDV, a hypertensive group consisting of women with hypertension and fetuses having EDV, and finally a small for gestational age group containing fetuses suspected of being small for gestational age but having EDV. If the mother was hypertensive and the fetus had EDV and was also suspected of being small for gestational age, that patient was allocated to the hypertensive group. These groups were created because each subset was managed differently, as explained later.

Doppler velocimetry was performed using continuous wave Doppler ultrasound (Doptek 9000, with 4 MHz transducer and wall filter at 100 Hz). If AEDV were detected, multiple other sites were examined to see if end–diastolic velocities could be found. If found, end diastolic flow was assumed to be present. Recordings were taken only when the fetus was apnoeic. Five waveforms were measured. The resistance index was calculated and plotted on reference curves already established for our population (Pattinson *et al.* 1989). A result was considered abnormal if it was greater than the 95th centile. In the study group, if AEDV were detected, the Doppler examination was repeated the following day to confirm AEDV and thereafter repeated weekly. These further results were also not revealed to the supervising clinician. In the control group the Doppler examination was repeated weekly if the woman was in hospital or fortnightly if she was managed as an outpatient. This was the standard procedure for any woman referred for Doppler examination.

Gestational age was assessed using the last normal menstrual period in conjunction with an ultrasound performed before 24 weeks gestation. If the dates differed from the ultrasound by more than two weeks, the ultrasound was used as the measure of gestational age. A small for gestational age baby was defined as a baby having a birthweight less than the 10th centile for the gestational age according to the growth curves of Yudkin *et al.* (1987).

Women who consented to the trial were randomised into the study group (in which the Doppler velocimetry was revealed), or a control group (in which the Doppler velocimetry result was withheld from the clinician). Randomisation was achieved using a balanced block technique such that at the completion of the block (groups of 10 for AEDV group and 20 for the other groups were used), there were equal numbers of women in each group. The allocation to study or control group was inserted into an opaque sealed envelope, and randomisation was performed by the person doing the Doppler velocimetry and after it had been completed.

If the patient was randomised to the study group, management guidelines were issued to the clinician managing the case. The AEDV group guidelines stated that if the fetus was between 28 and 30 weeks gestation or less than 1000 g, intensive fetal monitoring should be performed and delivery undertaken only if late decelerations were observed. If the fetus was 30 weeks gestation or more and more than 1000 g, early delivery should be considered.

For the group suspected of having a small for gestational age fetus, the guidelines stated that fetuses with normal resistance indices were not at risk and did not need intervention. The examination was repeated fortnightly. If the resistance index was abnormal but less than one, this indicated a possible fetal problem, and it was recommended that the Doppler examination be repeated weekly with twice weekly nonstress tests.

The guidelines for the hypertensive group stated that Doppler velocimetry could not predict abruption of the placenta and the hypertension must be managed according to its severity. Where indicated, intensive fetal monitoring was performed and the women managed according to our standard protocol (Odendaal *et al.* 1990). The control groups were managed by the clinician according to the clinical problem. Sonar and fetal heart rate monitoring was available to all patients.

The data were collected prospectively once the patient had been randomised. Data collection consisted of recording the following:

1. Maternal days in hospital if admission took place before labour or delivery;
2. Total days in hospital;
3. Maternal complications;
4. Induction of labour;
5. Method of delivery;
6. Perinatal deaths;
7. Admission to neonatal intensive care;
8. Neonatal morbidity;
9. Neonatal days in neonatal intensive care;
10. Neonatal days in hospital.

If the mother had a caesarean section, the infant was considered discharged from hospital if the baby was rooming in.

Calculation of sample size

Sample size was calculated as follows:

1. For fetuses between 28 and 32 weeks gestation with AEDV the perinatal mortality at our institution was 526/1000 deliveries (Pattinson *et al.* 1993). A clinically significant improvement in mortality was considered to be a decrease of the perinatal mortality to 200/1000 deliveries. It was estimated that 55 patients per limb would be required to have an 80% chance of demonstrating this at a level of significance of 0·05.
2. For the SGA group the intervention rate in mothers suspected of having small for gestational age fetuses was about 50% at Tygerberg Hospital. It was considered that halving this incidence would be clinically significant. Therefore, it was estimated that a sample size of 65 patients per limb was required to achieve a power of 80% at an error risk of 0·05. It was not anticipated that a difference would occur in the hypertension group so no prior hypothesis was stated.

Tygerberg Hospital is responsible for about 14000 deliveries per year, mainly from the lower socio-economic groups. However, it also serves as a tertiary referral hospital for the rural areas of the Cape Province which account for 20% of the births of less than 1500 g. The overall perinatal mortality (500 g to 28 days postpartum) is 34/1000 births. Neonatal intensive care facilities are available but because of the shortage of staff and facilities, they are usually restricted to babies weighing more than 1000 g.

Categorical data were analysed using the χ^2 test, odds ratio (OR) and 95% confidence limits (CL). Where the numbers were small, Fisher's exact test was used. Continuous variables were compared using Student's t test where the data were normally distributed and the Mann-Whitney U test for non-normal distributions. A significant difference was considered present if $P < 0.05$.

The trial was approved by the ethical committee of Tygerberg Hospital. Because of the nature of the trial, an undertaking was made to analyse the data at the completion of each block: if a significant difference between the groups was found, the trial would be stopped.

Results

A total of 212 women with pregnancies of 28 or more weeks gestation with hypertensive diseases and/or fetuses suspected of being small for gestational age entered the trial; no woman refused to participate in the trial. The number of patients from each group and the perinatal deaths are shown in Table 1.

Twenty women had fetuses with AEDV. The data concerning the study group are shown in Table 2 and those of the control group in Table 3. The mean maternal age of the study group was 25·8 years (SD 4·1) and that of the control group 29·8 years (SD 4·9). There were six primigravidas in the study group, and, of the remaining multiparas, two had had previous intrauterine deaths. The control group contained four primigravidas; three of the six multigravidas had had a previous intrauterine death. The gestational age and birthweight did not differ between the study group (30·7 weeks (SD 2·3) and 1123 g (SD 378 g)) and control group (30·3 weeks (SD 2·5) and 959 g (SD 252 g)). In the study group there was one neonatal death, whereas in the control group there were six deaths: five intrauterine and one perinatally related

Table 1. A comparison of perinatal deaths in the study and control groups. AEDV = absent end-diastolic velocities of the umbilical artery; SGA = small for gestational age; NS = no significant difference detected.

Group	Study *n*	Control *n*	*P*
AEDV	10	10	
Perinatal deaths	1	6	= 0·029
Hypertension	47	42	
Perinatal deaths	4	1	NS
Suspected SGA fetuses	51	52	
Perinatal deaths	1	0	NS
TOTAL	108	104	
Perinatal deaths	6	7	NS

infant death ($P = 0{\cdot}029$, Fisher's exact test). In none of the study group was there a reappearance of end-diastolic velocities.

Review of the data was carried out by the MRC Perinatal Mortality Unit after 20 cases as stipulated by the ethics committee. The trial was stopped at this point because significantly more fetuses had died in the control group. It had been previously agreed that if a clinically important difference between the groups was found during the regular interim analyses, the trial would be stopped. Once the randomisation in the AEDV group was stopped and all these results had to be revealed, the rest of the trial could not continue because the clinician could infer that any Doppler result not revealed would have to have end-diastolic velocity.

At the cessation of the trial a total of 103 women had

Table 2. Delivery and neonatal data of the AEDV study group.

No.	Pregnancy complications	Reason for delivery	Birthweight (g)	Gestation (wks)	NICU (days)	Outcome	Cause of death
1.	SGA	Placental abruption	760	30	9	Death	HMD
2.	SGA	Late dec	745	28	14	Live	—
3.	SGA	AEDV	1210	33	0	Live	—
4.	SGA	Late dec	780	29	30	Live	—
5.	SGA	AEDV	1220	35	0	Live	—
6.	SGA/HT/PP	Combined	1020	31	1	Live	—
7.	Pre-eclp	Combined	1035	28	69	Live	—
8.	Pre-eclp	AEDV	1980	33	5	Live	—
9.	Pre-eclp	AEDV	1460	30	0	Live	—
10.	Pre-eclp	Combined	1015	30	1	Live	—

NICU = neonatal intensive care; Pre-eclp = pre-eclampsia (defined as hypertension and proteinuria); HT = hypertension; SGA = small for gestational age; PP = placenta praevia; Late dec = late decelerations in the FHR pattern; AEDV = delivery due primarily to the presence of AEDV; Combined = Both maternal and fetal indications; HMD = hyaline membrane disease.

Table 3. Delivery and neonatal data of the AEDV control group.

No.	Pregnancy complications	Reason for delivery	Birthweight (g)	Gestation (wks)	NICU (days)	Outcome	Cause of death
1.	Proteinuria	IUD	500*	31	—	IUD	IUGR
2.	HT	Spont	700*	28	—	IUD	IUGR
3.	SGA	IUD	720*	30	—	IUD	IUGR
4.	SGA	IUD	1080*	36	—	IUD	IUGR
5.	SGA/epilepsy	IUD	960*	28	—	IUD	IUGR
6.	SGA	Late dec	1210	31	38	Death	NEC
7.	SGA/PP	Combined	885	28	40	Live	—
8.	Pre-eclp	Combined	950	29	12	Live	—
9.	Pre-eclp	Late dec	1340	32	—	Live	—
10.	Pre-eclp	Maternal	1190	30	—	Live	—

NICU = neonatal intensive care; Pre-eclp = pre-eclampsia (defined as hypertension and proteinuria); HT = hypertension; SGA = small for gestational age; PP = placenta praevia; Late dec = late decelerations; Combined = Both maternal and fetal indications; Spont = spontaneous onset; IUD = intrauterine death; NEC = necrotising enterocolitis; IUGR = intrauterine growth retardation.
* Macerated birthweights.

Table 4. Comparison of the characteristics between study and control group of women suspected of having SGA babies and having end–diastolic velocities. No significant differences detected between the groups.

	Study group ($n = 51$)	Control ($n = 53$)
Maternal age: years (SD)	26·5 (5·4)	26·4 (5·4)
Primigravidas: n (%)	16 (31%)	15 (28%)
Gestational age at entry: weeks (SD)	33·5 (2·6)	33·6 (2·7)
Pregnancy complications (developed after entry): n (%)		
Hypertension	2 (4%)	6 (12%)
Proteinuric hypertension	2 (4%)	2 (4%)
Spontaneous preterm labour	4 (8%)	4 (8%)
Antepartum haemorrhage	1 (2%)	2 (4%)
Light for gestational age babies	26 (51%)	28 (53%)

been entered in the small for gestational age part of the trial, 51 in the study group and 52 in the control group. The two groups were comparable with respect to the gestational age at entry to the study, maternal age, parity, complications developing after entry to the study, and number of babies who were subsequently classified as small for gestational age (Table 4). The outcome of the pregnancies with respect to the end points studied are shown in Table 5. The women in the study group had significantly fewer days in hospital before delivery ($P < 0·001$) and tended to have fewer maternal interventions (study group 27%, control group 43%; $P = 0·07$, OR 0·49, 95% CL 0·2 and 1·25), and caesarean sections (study group 13%, control group 27%; $P = 0·08$, OR 0·43, 85% CL 0·14 and 1·32). The infants of the study group in this subset also spent significantly less time in the neonatal wards ($P = 0·029$). There were no intrauterine deaths in this group. The neonatal death in the study group occurred in a woman who was assaulted and subsequently developed placental abruption.

Table 5. Comparison between the study group and control group of the subset with fetuses suspected of being SGA and having end–diastolic velocities with respect to end points. NS = no significant difference detected; NICU = neonatal intensive care unit.

	Study group ($n = 51$)	Control group ($n = 53$)	P
Gestational age at delivery: weeks (SD)	37·6 (2·6)	37·3 (2·5)	NS
Birthweight: g (SD)	2477 (539)	2364 (639)	NS
Hospitalisation days: median (1st and 3rd quartile)			
Maternal antenatal	0 (0–1)	2 (0–5)	< 0·05
Neonatal	1·5 (1–6)	4 (1–7)	< 0·05
Spontaneous labour: n (%)	38 (75%)	28 (53%)	= 0·07
Caesarean sections: n (%)	7 (14%)	13 (25%)	= 0·08
Antenatal fetal distress: n (%)	0	6 (9%)	< 0·05
NICU admissions: n (%)	0	5 (9%)	< 0·05
Deaths: n (%)	1 (2%)	1 (2%)	NS

Table 6. Comparison between the study group and control group of the subset with hypertension where the fetuses had end–diastolic velocities with respect to end points. NS = no significant difference detected; NICU = neonatal intensive care unit.

	Study group ($n = 47$)	Control group ($n = 42$)	P
Gestational age			
At entry: weeks (SD)	31·9 (2·4)	31·8 (2·3)	NS
At delivery: weeks (SD)	34·3 (3·1)	33·7 (3·3)	NS
Birthweight: g (SD)	2015 (775)	1916 (648)	NS
Hospitalisation days: median (1st and 3rd quartile)			
Maternal antenatal	5 (2–6)	5·5 (1·5–6)	NS
Neonatal	7 (5–11)	6 (2–10·5)	NS
Spontaneous labour: n (%)	8 (17%)	4 (10%)	NS
Caesarean sections: n (%)	27 (57%)	23 (55%)	NS
Antenatal fetal distress: n (%)	2 (4%)	1 (2%)	NS
NICU admissions: n (%)	12 (26%)	11 (26%)	NS
Deaths: n (%)	4 (9%)	1 (2%)	NS

The results of the hypertension group are shown in Table 6. There were no differences between the study and control groups in this subset. There were no intrauterine deaths in this group. The deaths of the infants in the study group were due to grade IV hyaline membrane disease (2), septicaemia, and necrotising enterocolitis. The infant death in the control group was due to septicaemia.

Discussion

Three randomised controlled trials testing the value of Doppler velocimetry in the management of pregnancies have been reported (Trudinger *et al.* 1987; Tyrrell *et al.* 1990; Newnham *et al.* 1991). All interpreted their findings as showing that knowledge of the Doppler velocimetry result might improve the management but any improvement was slight and often not clinically significant. In all these studies, the effect of Doppler velocimetry as a whole was analysed. The presence of AEDV implies impending fetal distress and is associated with a high perinatal mortality and morbidity. Conversely, the presence of EDV implies adequate placental reserve. Consequently, the management of fetuses with AEDV will differ radically from those with EDV. This confounds the interpretation of trials in which the value of Doppler velocimetry as a whole is assessed because the end points are mixed. For example, when the Doppler indices are increased more elective deliveries may be indicated, whereas when the Doppler indices are normal women may be allowed to await spontaneous labour. To test whether Doppler velocimetry is of value, the effect of the knowledge of the Doppler result in a specific set of circumstances needs to be compared with the standard management of women in the same set of circumstances.

In this randomised controlled trial, knowledge that the fetus had AEDV significantly improved the outcome of those fetuses. It is possible that the two groups are not entirely comparable because the birthweights of the babies in the control group were lower although not significantly so. This trend might be explained by the five intrauterine deaths (which were all macerated) in the control group. It is uncertain what happens to the birthweight of babies who die *in utero*; perhaps they lose weight. Also the maternal ages tended to be lower in the study group but otherwise in parity, previous obstetric histories and pregnancy complications the groups were comparable.

By giving the responsible clinician a management guideline for a fetus with AEDV, we might have biased the outcome because the clinician was aware we were specifically interested in the outcome and so more care might have been taken. However, by referring the woman for Doppler velocimetry the clinician knew we were interested in the outcome of the pregnancy.

The guideline might have helped in another way, especially for a clinician not specifically involved in high risk obstetrics, in that he/she would have had a specific management plan to follow. This was not the case where the Doppler findings were withheld. Consultants in the department of obstetrics and gynaecology at Tygerberg Hospital are responsible for obstetric patients even if their major interest is in the field of infertility or gynaecological oncology. Therefore, it is possible that these results for the control group would not be repeated in units where all high risk pregnancies are managed by clinicians who are subspecialists in perinatal medicine. On the other hand, this strengthens the argument for the use of Doppler velocimetry in circumstances such as those prevailing at Tygerberg Hospital, because the report of AEDV specifically identifies fetuses needing more attention.

This study does not contribute to solving the problem of whether fetuses with AEDV should be delivered at a specific gestation or estimated fetal weight because the study was stopped prematurely.

Small for gestational age babies are most commonly small normal babies, but some are growth retarded. There are various causes of intrauterine growth retardation; an important one is placental insufficiency. Doppler velocimetry of the umbilical artery gives a measure of placental resistance which is usually abnormal in placental insufficiency. This suggests that Doppler velocimetry can often distinguish between the small fetus that can safely be managed conservatively from the fetus at high risk of developing fetal distress or perinatal death who is likely to benefit from earlier delivery. This study shows that women who are suspected of having small for gestational age fetuses but where EDV are present can be managed more cost effectively by using Doppler velocimetry.

In our population the majority of fetuses with AEDV come from women with hypertensive diseases. Fetuses of mothers with hypertension and having EDV had to be included in this study because knowledge of the Doppler velocimetry result in this group might have influenced the clinicians' decision making adversely, and this may have cancelled out the advantage gained with the knowledge that AEDV was present. This did not happen in this study, suggesting that knowing the Doppler result in women with hypertension was beneficial.

Within the confines of this study, knowledge of the Doppler velocimetry result was beneficial in the subsets where AEDV was present, and where small for gestational age fetuses were suspected but EDV was present. In the subset in which women had hypertension but whose fetuses had EDV, there was no beneficial or adverse effect. Therefore, in clinical settings such as ours Doppler velocimetry has an important role in the management of high risk pregnancies along with fetal heart rate monitoring and real time ultrasound.

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References

Burke G., Stuart B., Crowley P., Scanaill S. & Drumm J. (1990) Is intrauterine growth retardation with normal umbilical artery blood flow a benign condition? *Br Med J* **300**, 1044–1045.

Haddad N. G., Johnstone F. D., Hoskins P. R., Chambers S. E., Muir B. B. & McDicken W. N. (1988) Umbilical artery Doppler waveform in pregnancies with uncomplicated intrauterine growth retardation. *Gynecol Obstet Invest* **26**, 206–210.

Low J. A. (1991) The current status of maternal and fetal blood flow velocimetry (National Institute of Health Consensus Conference 1986). *Am J Obstet Gynecol* **164**, 1049–1063.

McParland P., Steel S. & Pearce J. M. (1990) The clinical implications of absent or reversed end-diastolic frequencies in the umbilical artery flow velocity waveforms. *Eur J Obstet Gynecol Reprod Biol* **37**, 15–23.

Neilson J. P. (1987) Doppler ultrasound. *Br J Obstet Gynaecol* **94**, 929–934.

Newnham J. P., O'Dea M. R.-A., Reid K. P. & Diepeveen D. A. (1991) Doppler flow velocity waveform analysis in high risk pregnancies: a randomized controlled trial. *Br J Obstet Gynaecol* **98**, 956–963.

Odendaal H. J., Pattinson R. C., Bam R., Grove D. & Kotze T. J. (1990) Aggressive or expectant management for patients with severe pre-eclampsia between 28 and 34 weeks gestation: a randomised controlled trial. *Obstet Gynecol* **76**, 1070–1075.

Pattinson R. C. (1992) The perinatal outcome of antenatally diagnosed small for gestational age fetuses who have normal Doppler indices of the umbilical artery. In M.D. Thesis, University of Stellenbosch: The role of Doppler velocimetry of the umbilical artery in assessing fetal health, pp. 131–143.

Pattinson R. C., Dawes G., Jennings J. & Redman C. W. G. (1991) Umbilical artery resistance index as a screening test for fetal well-being. I: Prospective revealed evaluation. *Obstet Gynecol* **78**, 353–358.

Pattinson R. C., Odendaal H. J. & Kirsten G. (1993) The relationship between absent end-diastolic velocities of the umbilical artery and perinatal mortality, neonatal morbidity and long term outcome. *Early Hum Dev* **33**, 61–69.

Rochelson B. L., Schulman H., Fleischer A. *et al.* (1987) The clinical significance of Doppler umbilical artery velocimetry in the small for gestational age fetus. *Am J Obstet Gynecol* **156**, 1223–1226.

Trudinger B. J., Giles W. B., Cook C. M., Connelly A. & Thompson R. S. (1987) Umbilical artery flow velocity waveforms in high-risk pregnancy: randomised controlled trial. *Lancet* **i**, 188–190.

Tyrrell S. N., Lilford R. J., MacDonald H. N., Nelson E. J., Porter J. & Gupta J. K. (1990) Randomized comparison of routine vs highly selective use of Doppler ultrasound and biophysical scoring to investigate high risk pregnancies. *Br J Obstet Gynaecol* **97**, 909–916.

Yudkin P. L., Aboualfa J. A. E., Redman C. W. G. & Wilkinson A. R. (1987) New birthweight and head circumference centiles for gestational ages 24 to 42 weeks. *Early Hum Dev* **15**, 45–52.

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